



ME 200

Thermodynamics

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Lecture 04 – 1st Law of Thermodynamics

Cooling a Gas in a Piston–Cylinder

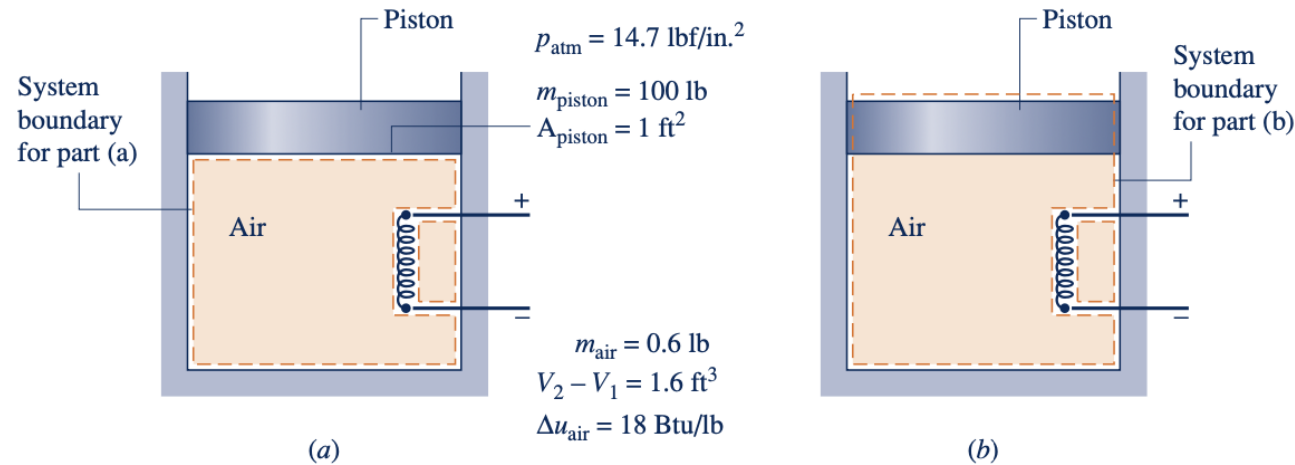
Four-tenths kilogram of a certain gas is contained within a piston–cylinder assembly. The gas undergoes a process for which the pressure–volume relationship is

$$pV^{1.5} = \text{constant}$$

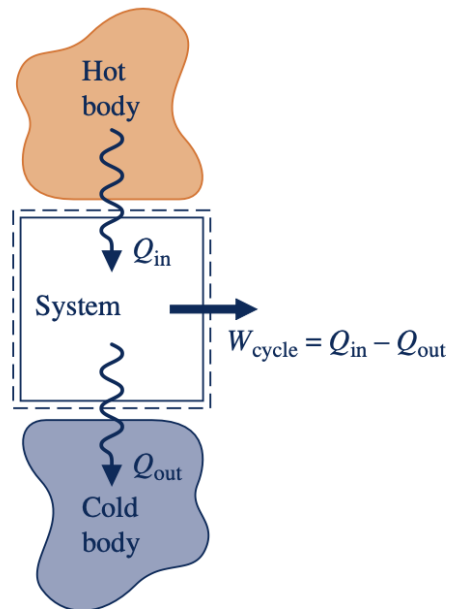
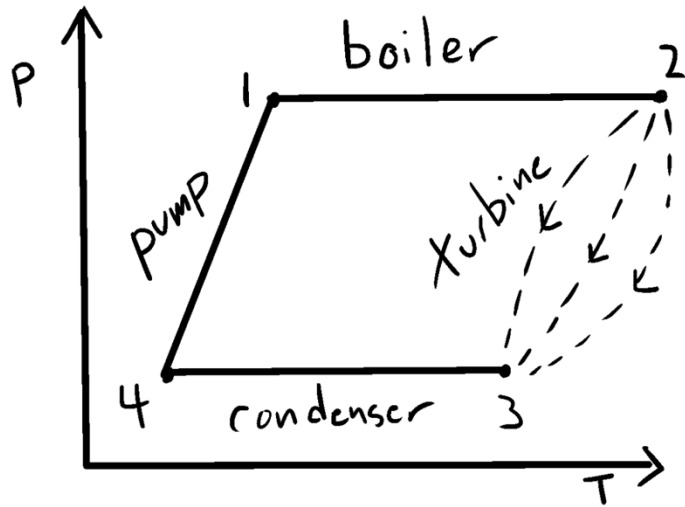
The initial pressure is 3 bar, the initial volume is 0.1 m^3 , and the final volume is 0.2 m^3 . The change in specific internal energy of the gas in the process is $u_2 - u_1 = -55 \text{ kJ/kg}$. There are no significant changes in kinetic or potential energy. Determine the net heat transfer for the process, in kJ.

Air is contained in a vertical piston–cylinder assembly fitted with an electrical resistor. The atmosphere exerts a pressure of 14.7 lbf/in.^2 on the top of the piston, which has a mass of 100 lb and a face area of 1 ft^2 . Electric current passes through the resistor, and the volume of the air slowly increases by 1.6 ft^3 while its pressure remains constant. The mass of the air is 0.6 lb , and its specific internal energy increases by 18 Btu/lb . The air and piston are at rest initially and finally. The piston–cylinder material is a ceramic composite and thus a good insulator. Friction between the piston and cylinder wall can be ignored, and the local acceleration of gravity is $g = 32.0 \text{ ft/s}^2$. Determine the heat transfer from the resistor to the air, in Btu, for a system consisting of **(a)** the air alone, **(b)** the air and the piston.

Schematic and Given Data:



Energy Analysis of Cycles



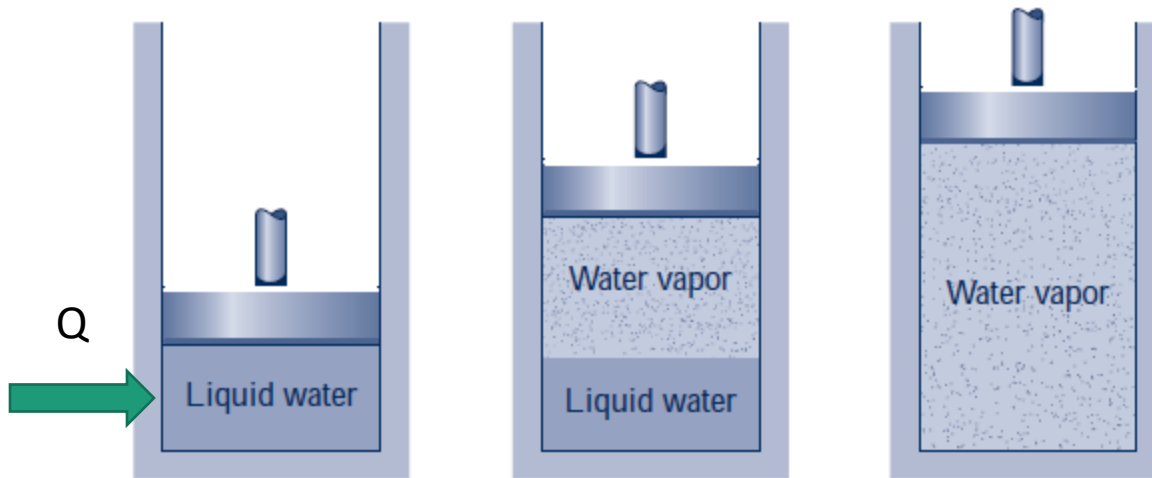
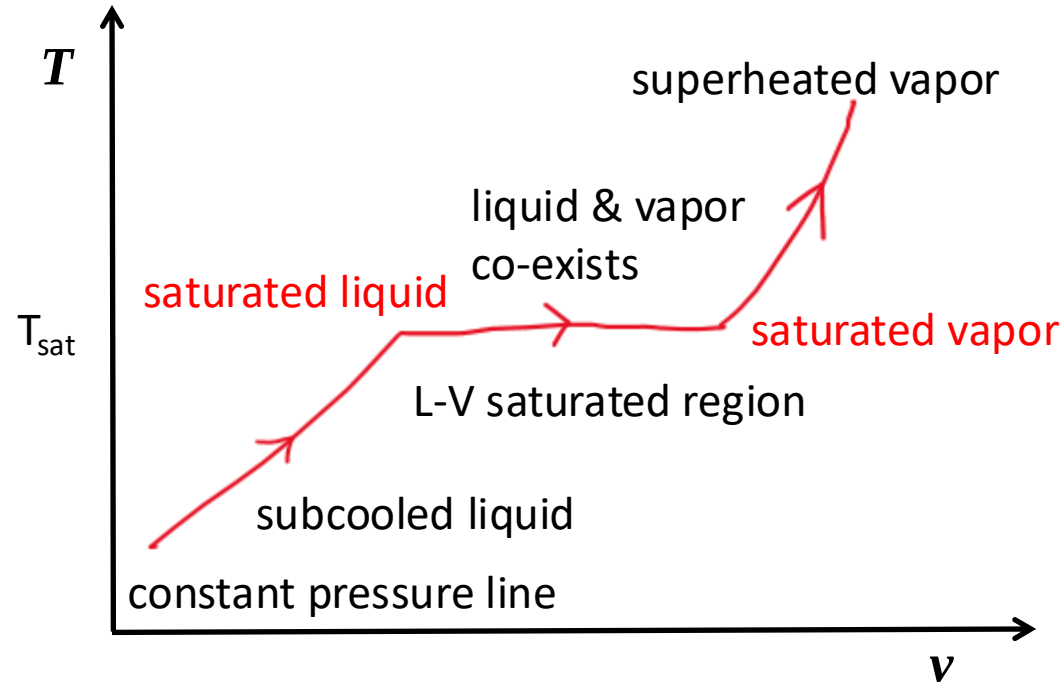
thermal efficiency

p - v - T Relation

Isobaric expansion:

$$p = \frac{\text{weight}}{A} = \text{constant pressure}$$

$$pv = RT$$



$$\frac{Q}{m} = \frac{W}{m} + u_{f_{\text{final}}} - u_{i_{\text{initial}}}$$

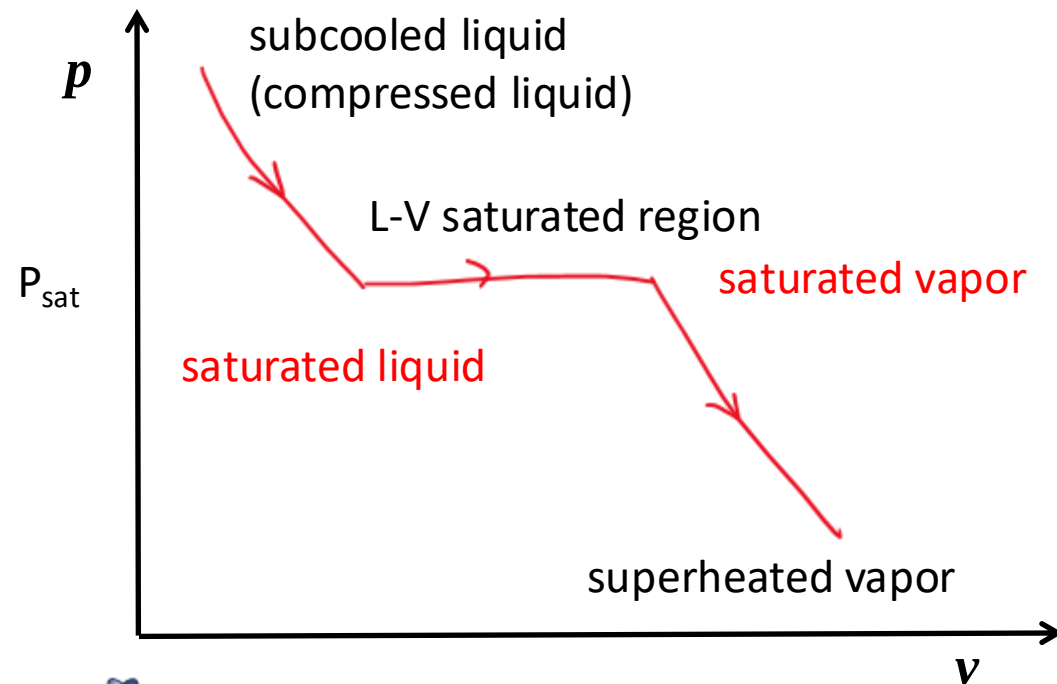
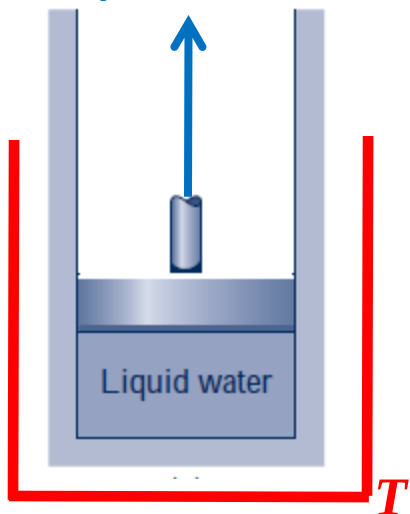
$$\frac{W}{m} = p(v_{\text{final}} - v_{i_{\text{initial}}})$$

p - v - T Relation

Isothermal expansion:

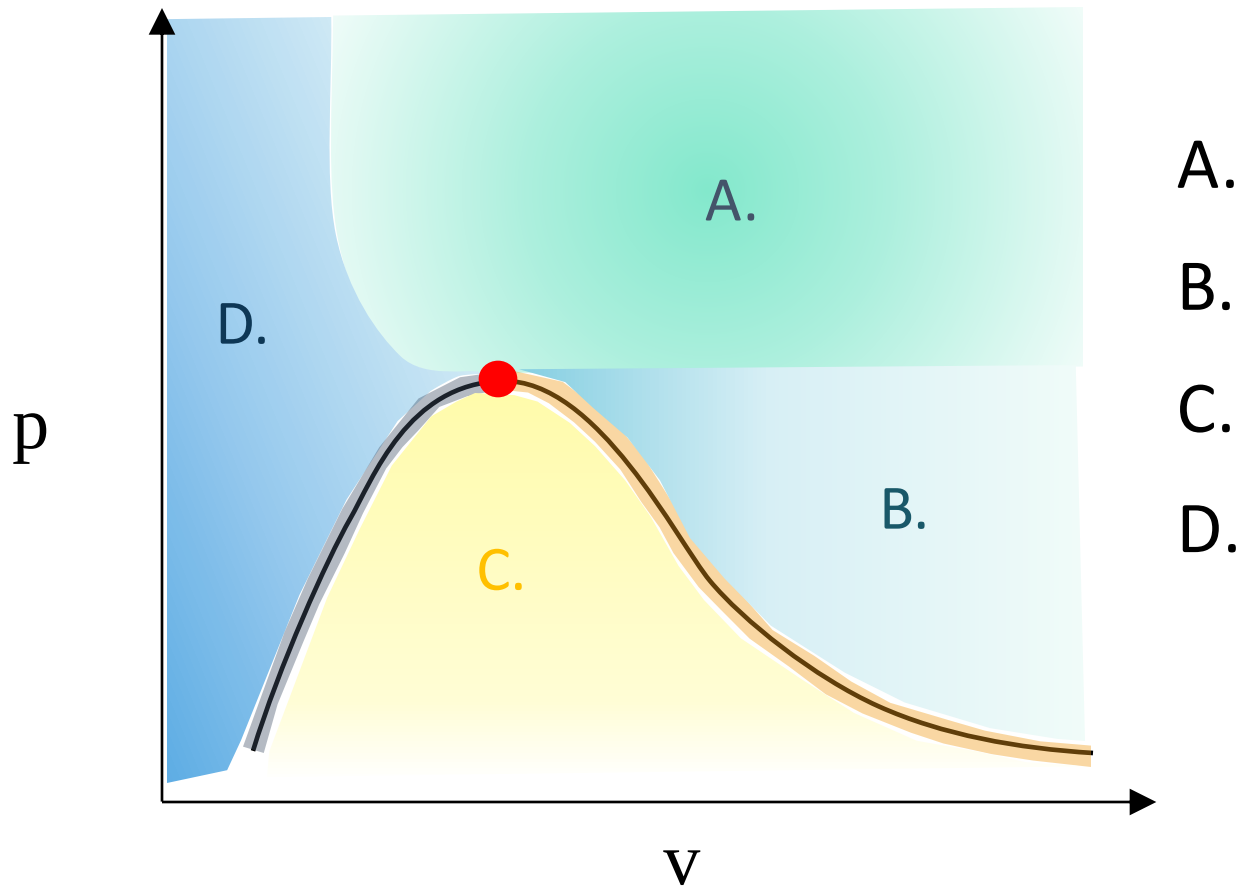
$$Q > 0$$

reducing
pressure



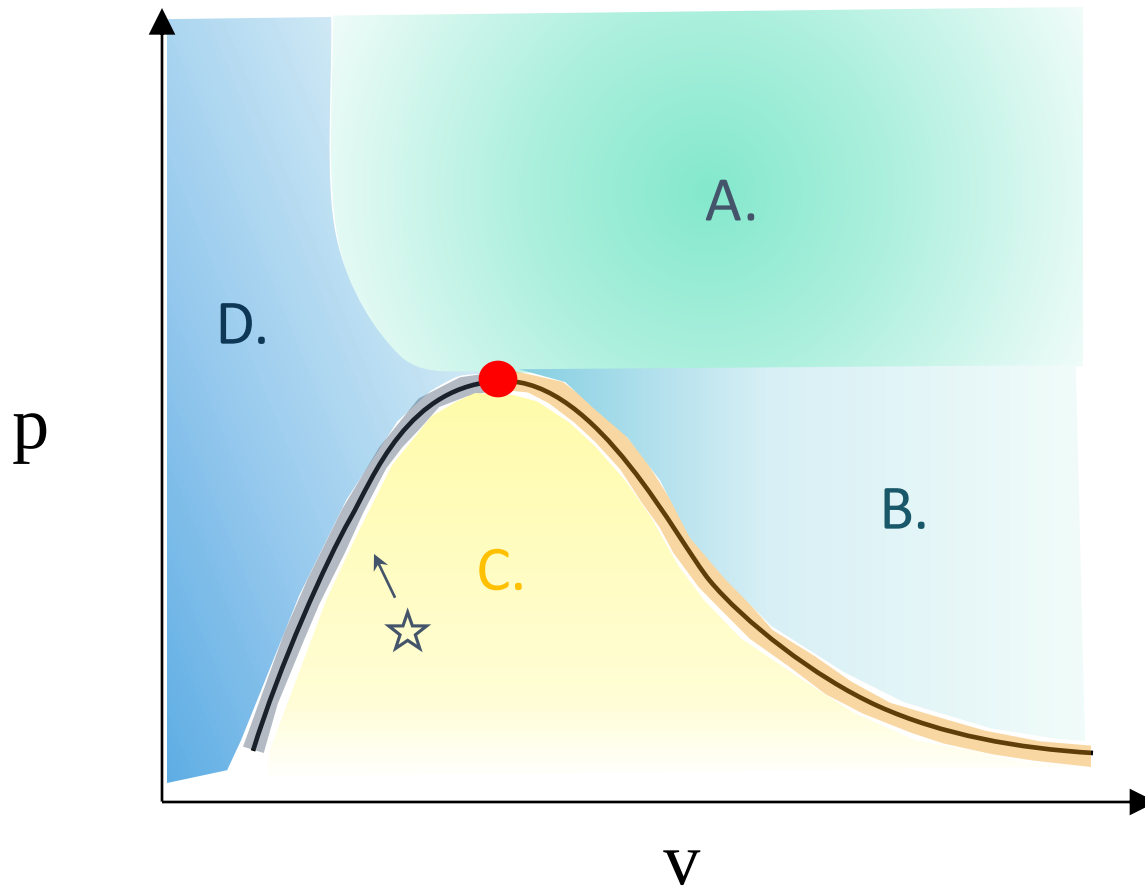
p - v Diagram

A pure substance in the liquid state will be found in this region...



p - v Diagram

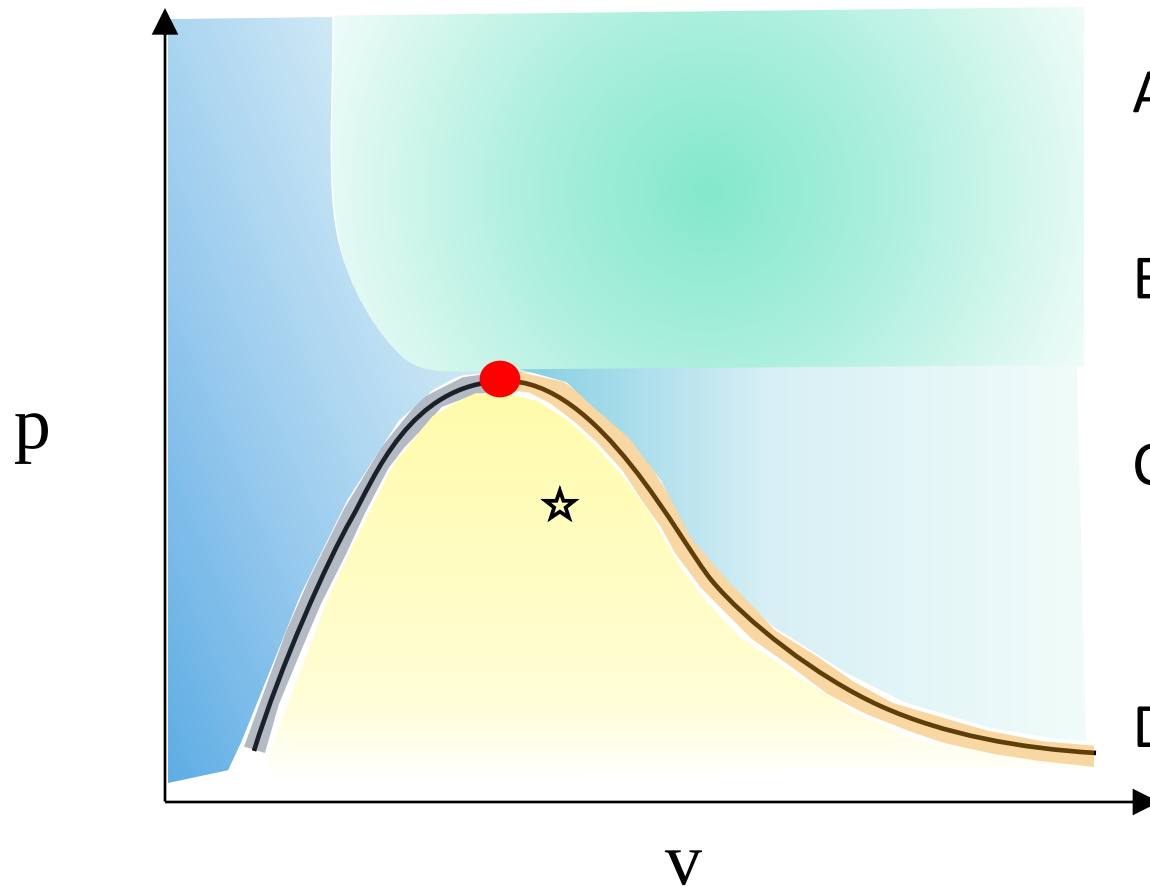
If the state of a substance lies on the line indicated, it is called a...



- A. Subcooled liquid
- B. Two-phase L+V
- C. Saturated vapor
- D. Saturated liquid

p - v Diagram

If the state of a substance lies at the point indicated, which of the following is true?

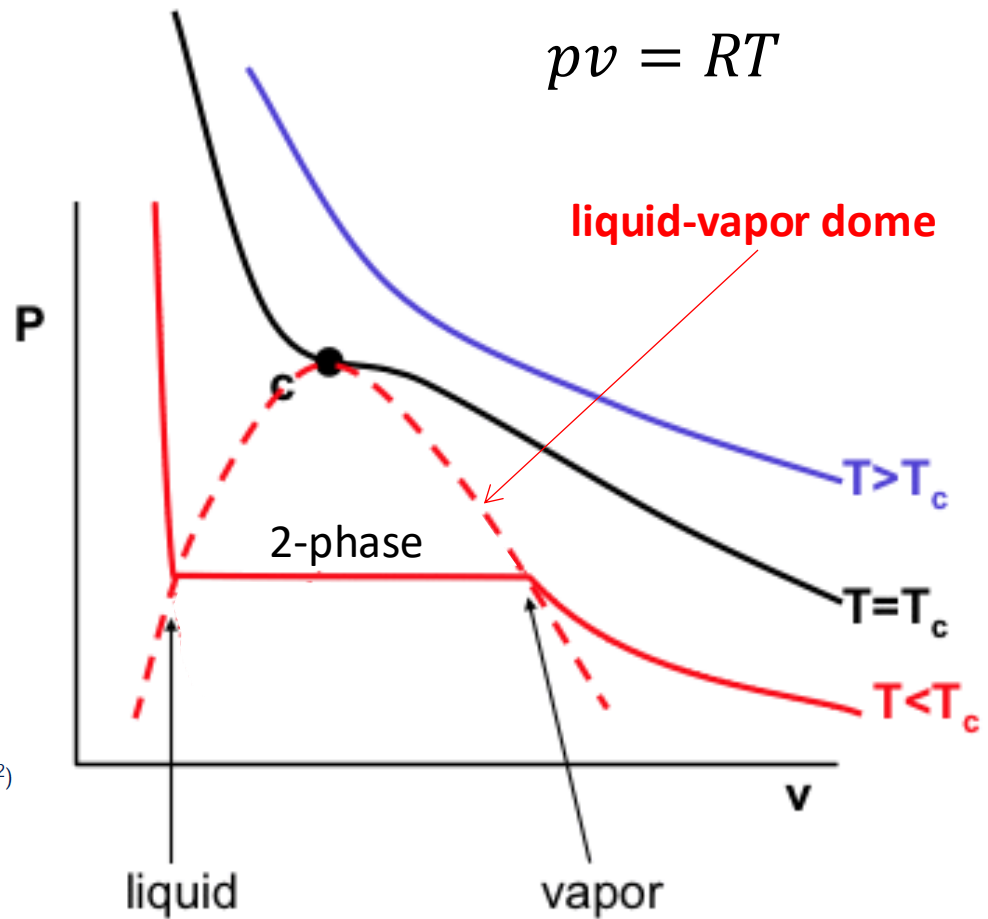
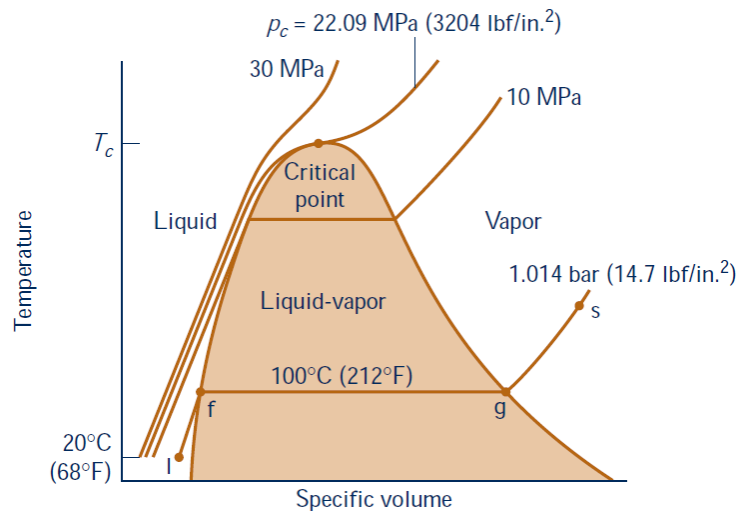


- A. There is more vapor than liquid.
- B. There is more liquid than vapor
- C. There are about equal amounts liquid and vapor.
- D. Don't know

p - v Diagram

During phase changes, p & T
are not independent!

Define: Quality



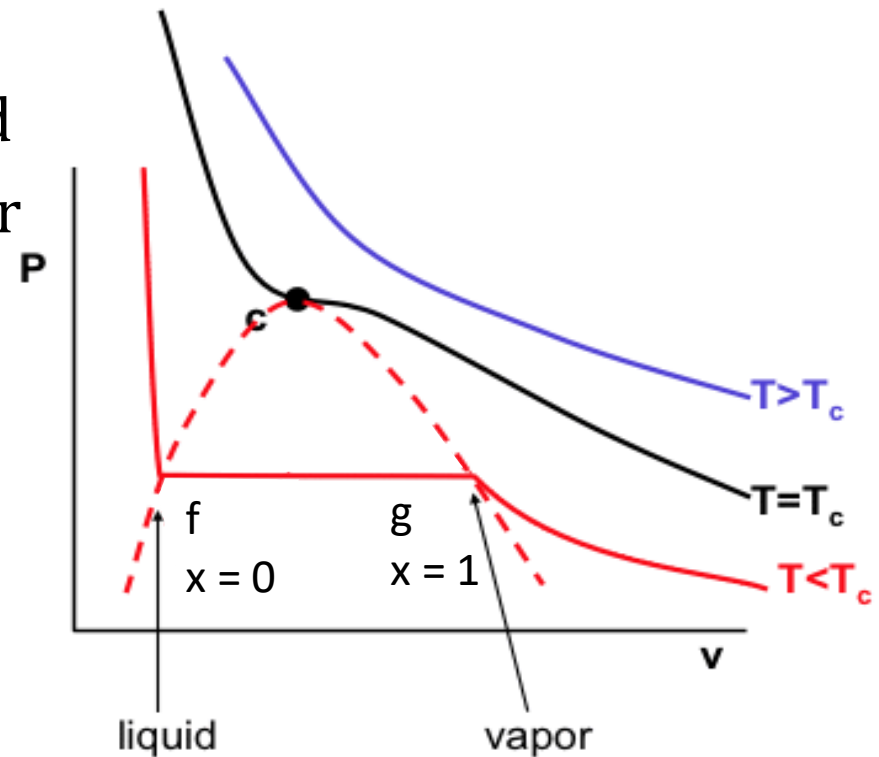
Quality

Define: Quality x - Vapor mass ratio

$$x = \frac{m_{\text{vapor}}}{m_{\text{liquid}} + m_{\text{vapor}}} = \frac{m_{\text{vapor}}}{m_{\text{total}}}$$

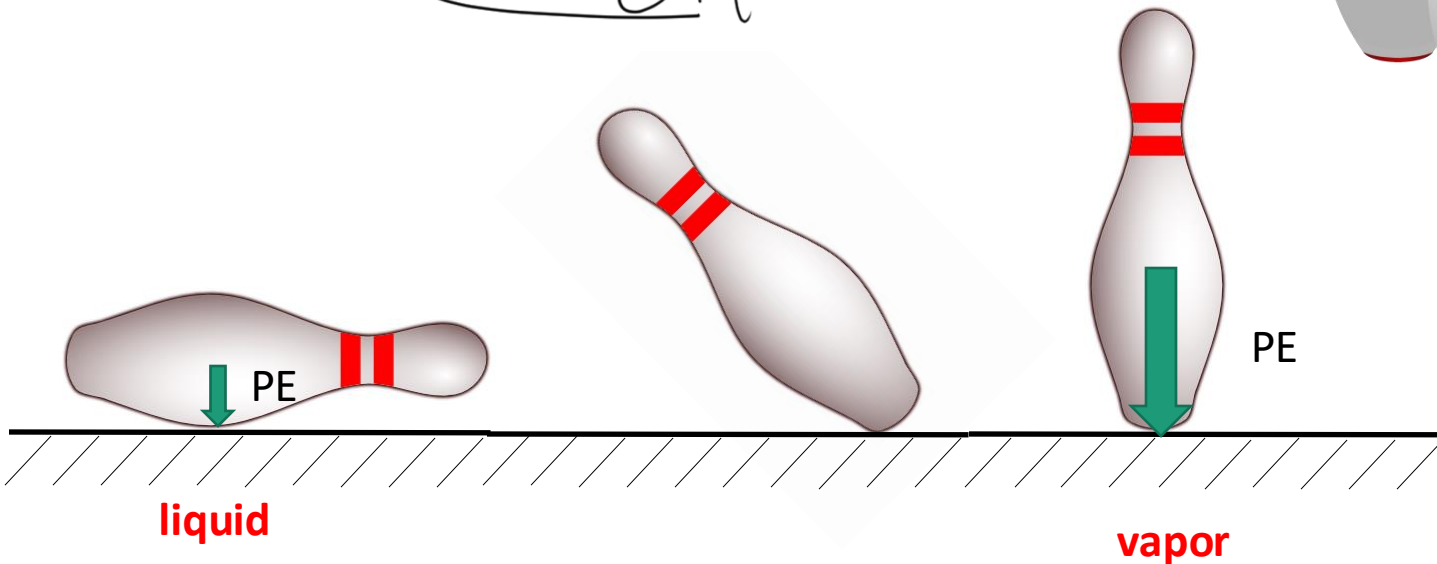
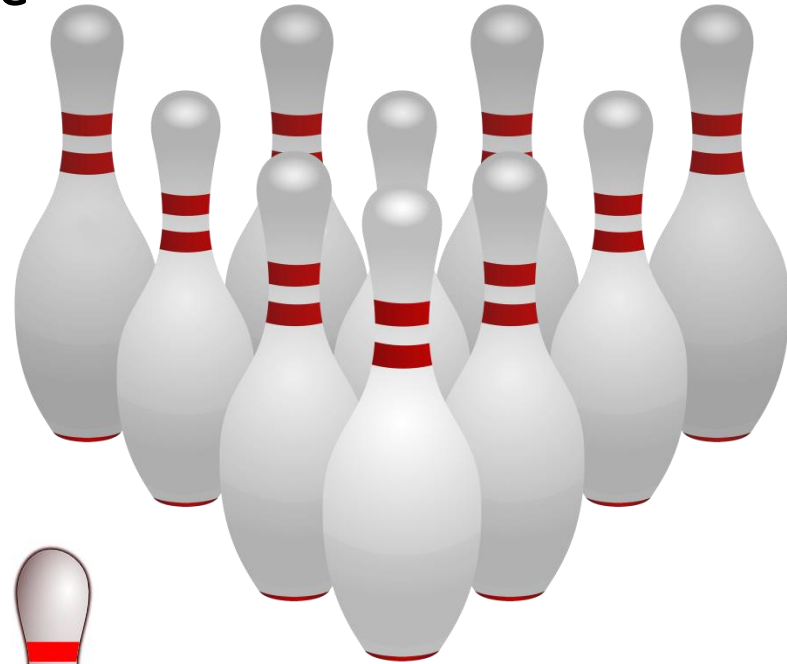
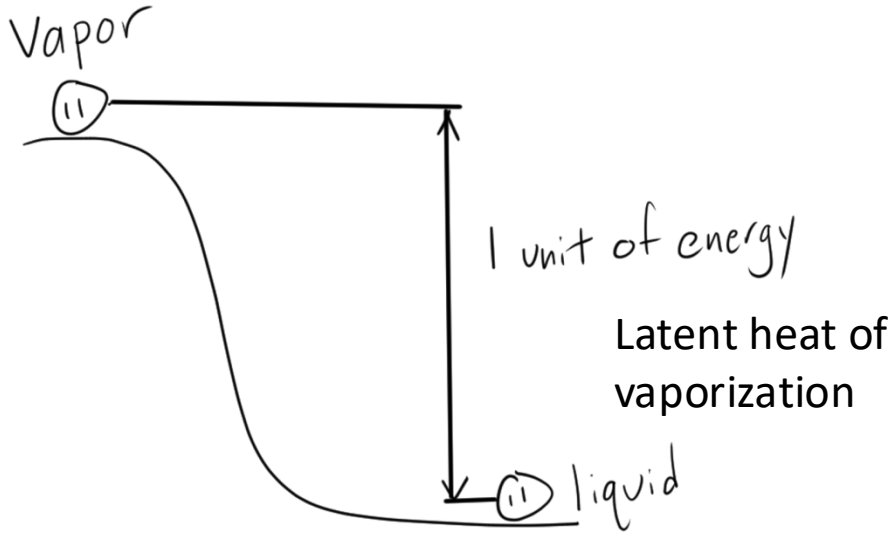
$x = 0$: v_f, u_f, h_f for saturated liquid

$x = 1$: v_g, u_g, h_g for saturated vapor



Quality

Bowling Pin Analogy to a Change of Phase



Property Calculation in Liquid-Vapor Dome

Bowling Pin Analogy to a Change of Phase

- Quality x – vapor-mass fraction

$$v = \left(\frac{m_{\text{liquid}}}{m}\right) v_f + \left(\frac{m_{\text{vapor}}}{m}\right) v_g$$

$$v = (1 - x)v_f + xv_g$$

$$= v_f + x(v_g - v_f)$$

Can find u , h , s

